Roadmap. Srednicki §11: how LecNote9's amplitude $\mathcal{T}$ becomes an observable. Box normalization turns $|\langle f|i\rangle|^2$ into the relativistic golden rule, $d\sigma = |\mathcal{T}|^2\, d\mathrm{LIPS}/(4|\mathbf{k}_1^{\mathrm{CM}}|\sqrt{s})$ and $d\Gamma = |\mathcal{T}|^2\, d\mathrm{LIPS}/2E_1$, then $d\sigma/d\Omega$ and $d\sigma/dt$; pp. 12–14 add barns, luminosity and PDG phase space. Used for $\mathcal{T}$ at one loop (LecNote14, §20), $|\mathcal{T}_{\mathrm{obs}}|^2$ (LecNote15, §26), and the Yukawa and QED cross sections (LecNote20, §48; LecNote23, §59). Supplements: CrossSection_GoldenRule, CrossSection_Lorentz, CrossSection_WorkedExample.

(1)

Section 11

Cross sections & decay rates

In most cases, we only deal with two kinds of cases:

① one particle initial state

(Decay) Decay rate.

② two particle initial state

(scattering) Cross section.

Cross section is in units of Length^2 (motivate).

Previously, we got that, the matrix element.

$$\mathcal{T} = g^2\left[\frac{1}{(k_1+k_2)^2+m^2} + \frac{1}{(k_1-k_1')^2+m^2} + \frac{1}{(k_2-k_2')^2+m^2}\right] + O(g^4). \quad (1)$$

k_1 and k_2 are the four-momenta of the two incoming particles

k_1' and k_2' are the four-momenta of the outgoing particles

1 Should read $(k_1 - k_2')^2$ in the third term, the u-channel: $k_2 - k_2' = k_1' - k_1$, so the term as written repeats the t-channel. As on LecNote9 p. 8: $\mathcal{T} = g^2\left[\frac{1}{(k_1+k_2)^2+m^2} + \frac{1}{(k_1-k_1')^2+m^2} + \frac{1}{(k_1-k_2')^2+m^2}\right] + O(g^4)$.

Energy Momentum is conserved

$$k_1 + k_2 = k_1' + k_2'$$

We also set all particles on-shell.

$$k_i^2 = -m_i^2$$

↑ for general use, we allow for different masses.

We work out some kinematics.

Define the center of mass (CM) frame. $\vec{k}_1 + \vec{k}_2 = 0 = \vec{k}_{1'} + \vec{k}_{2'}$

Choose $\vec{k}_1$ in the positive z direction, only $|\vec{k}_1| \equiv k_z$ is left unspecified.

The total energy is

$$E_1 + E_2 = \sqrt{|\vec{k}_1|^2 + m_1^2} + \sqrt{|-\vec{k}_1|^2 + m_2^2}$$

In fact, it is more convenient to define a L.I. scalar product.

$$s = -(k_1 + k_2)^2 \quad \text{which is } (E_1 + E_2)^2 \text{ in the CM frame.}$$

s is call the center of mass energy squared.

$$\underbrace{|\vec{k}_1|}_{\text{not L.I.}} = \frac{1}{2\sqrt{s}} \sqrt{s^2 - 2(m_1^2 + m_2^2)s + (m_1^2 - m_2^2)^2} \qquad \text{(CM frame)}$$

Similarly, we also have in the final state

$$|\vec{k}_{1,1}| = \frac{1}{2\sqrt{s}}\sqrt{s^2 - 2(m_{1,1}^2 + m_{2,1}^2)s + (m_{1,1}^2 - m_{2,1}^2)^2}$$

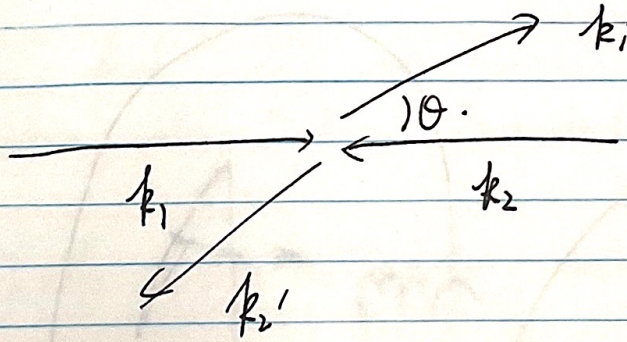
The total d.o.f is 6 for 2-unspecific on-shell final state particles,

$6 - \underline{4} = 2$, which can be identified as polar & Arthimutal angles

Energy-momentum conservation

The arthimutal angle is typically irrelevant as the system would be rotationally (along +z axis) invariant.

So the remaining most physical d.o.f is the polar angle θ (in the CM frame).



beyond $s \equiv -(k_1 + k_2)^2$

there are two other Lorentz Invariant Mandelstam variables

$$t = -(k_1 - k_1')^2 = -k_1^2 - k_{1,1}^2 + 2k_1 \cdot k_2$$
$$= m_1^2 + m_{1,1}^2 - 2E_1 E_{1,1} + 2|\vec{k}_1||\vec{k}_{1,1}|\cos\theta.$$

~~$u = -(k_1 - k_2')^2$~~

$$t = -(k_1 - k_1')^2 = -(k_2 - k_2')^2$$

and $u \equiv -(k_1 - k_{2,1})^2 = -(k_2 - k_{1,1})^2$

These three Mandelstein variables are not linearly independent, but they are quite useful in many occasions.

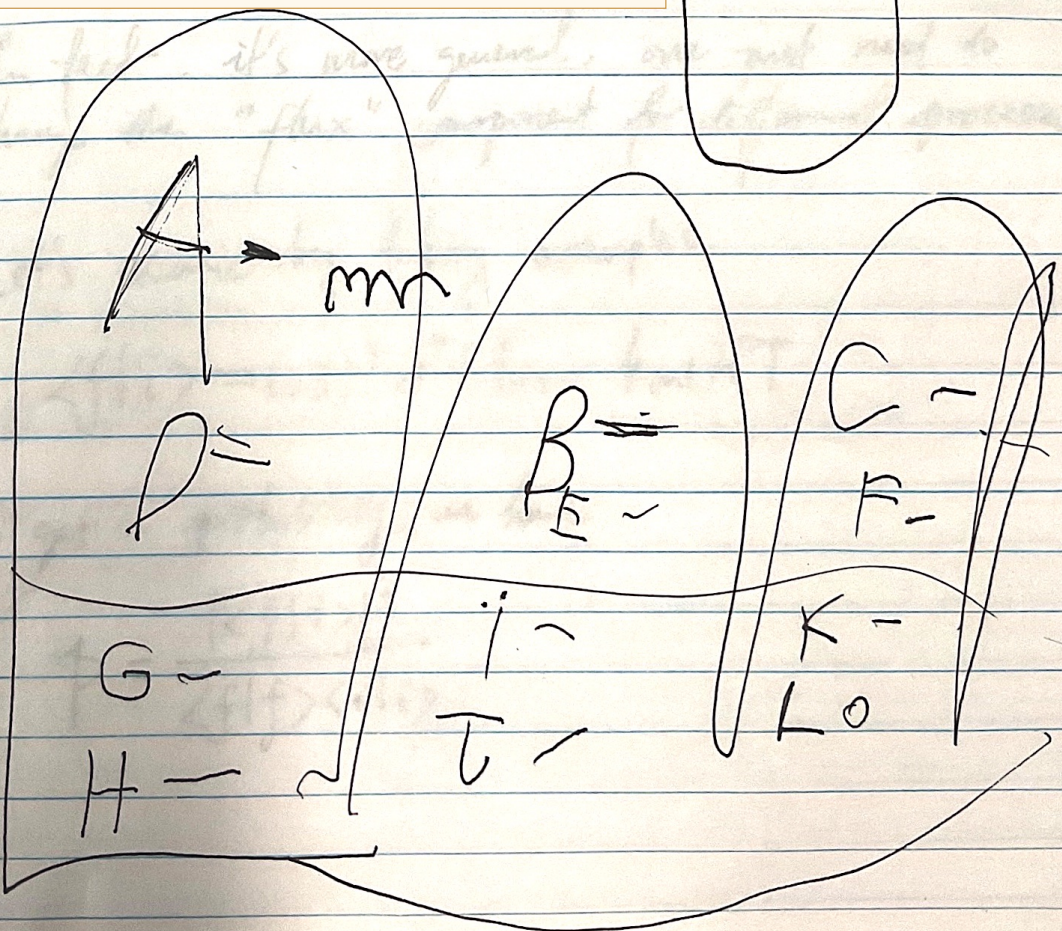
They can make certain symmetry properties of the amplitudes manifest.

$$s + t + u = m_1^2 + m_2^2 + m_{1'}^2 + m_{2'}^2$$

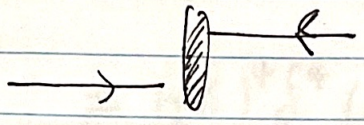
so

$$\mathcal{T} = g^2 \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right] + \mathcal{O}(g^4) \quad (1)$$

1 Should read $\mathcal{T} = g^2[1/(m^2 - s) + 1/(m^2 - t) + 1/(m^2 - u)] + O(g^4)$, since $(k_1 + k_2)^2 + m^2 = m^2 - s$ (LecNote9 p. 8). For $s \geq 4m^2$, $t, u \leq 0$ no denominator vanishes, so the $i\epsilon$ can be dropped.



Now, introducing the concept of cross-section.



"geometric" size normal to the incident direction

(If I throw two balls to each other, what's their chance to scatter?)(The ball spread $\propto$ De Broglie wavelength, 用)

This is an observable by colliding two beams of particles and check out the scattering probability. → for $2 \to n'$, $\frac{1}{\text{flux}}$

$$d\sigma = \frac{1}{\text{flux}} |T|^2 \, d\text{LIPS}_{n'}$$

→ for $1 \to n'$, $\frac{1}{2E_1}$ [1]

This is the relativistic version of the Fermi-golden rule for scattering.

1 "Flux" here is the flux factor, not p. 8's flux. For $2 \to n'$ it is $4|\mathbf{k}_1^{\text{CM}}|\sqrt{s}$ (p. 9): p. 8's $|\mathbf{k}_1|/(E_1 V)$ times the $4E_1E_2V$ in p. 7's $\dot{P}$, a mass^2. For $1 \to n'$ it is $2E_1$ (p. 10): $\langle i|i\rangle = 2E_1V$ already cancels the V of VT, so no flux is needed.

In fact, it's more general, one just need to change the "flux" component for different processes.

Let's examine by taking examples.

$$\langle f|i\rangle = (2\pi)^4 \delta^4(k_{in} - k_{out}) i\mathcal{T}$$

To get a probability, we have

$$\mathcal{P} = \frac{|\langle f|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle}$$

1 Why $\delta^4(0)$? Momentum eigenstates are plane waves filling all space and time, so $|\langle f|i\rangle|^2$ carries their overlap $(2\pi)^4\delta^4(0) = VT$. They are not unit-normalized (box: $\langle k|k\rangle = 2k^0V$), hence $\langle i|i\rangle\langle f|f\rangle$ and the $1/2E$'s. Net $P \propto T/V$: a constant rate, and the $1/V$ (one beam particle per box) is what the flux removes (p. 8).

$$\langle f|i\rangle^2 = (2\pi)^4\delta^4\left(\sum k_{in} - k_{out}\right)^2 |T|^2$$

$$= (2\pi)^4\delta^4(k_{in} - k_{out}) \cdot (2\pi)^4\delta^4(0)\,|T|^2$$

$$(2\pi)^4\delta^4(0) = \int d^4x\, e^{i0x} = VT \quad (1)$$

$\to$ volume of space $\times$ volume of time.

one-particle canonical normalization

$$\langle k|k\rangle = (2\pi)^3 2k^0\delta^3(\vec{0}) = 2k^0V$$

so for two particle initial state

$$\langle i|i\rangle = 4E_1E_2V.$$

$$\langle f|f\rangle = \frac{n!}{\prod_{j=1}^{n}} 2k_j^0 V$$

but $\langle f|i\rangle$ is the amplitude for a fixed final state momentum, we need to account for measurable quantities (small window of the continuous variables).

Within a finite sized box, a particle's momentum is quantized

$$\vec{k}_j' = \left(\frac{2\pi}{L}\right)\vec{n}_j \quad , \quad \vec{n}_j \text{ is an integer vector.}$$

in limit of large $L = (V)^{1/3}$

$$\sum_{n_j} \to \frac{V}{(2\pi)^3}\int d^3\vec{k}_j'$$

The quantized momentum goes to the continuous limit.

$$\int \frac{V}{(2\pi)^3} d^3\vec{k}_j = \sum_{n_j}\left(\frac{2\pi}{L}\right)^3 \vec{n}_j \Rightarrow$$

~~$d^3\vec{k}_j$~~

So for each outgoing particle. to get a probability per unit time

~~$d\Pi = (2\pi)^3$~~

P.S. → ~~$\frac{(2\pi)^3}{V}$~~ $\frac{V}{(2\pi)^3} d^3\vec{k}_j$

$$\dot{P} = \frac{(2\pi)^4 \delta^4(k_{in} - k_{out})}{4E_1 E_2 V} |T|^2 \frac{n'}{\prod_{j=1}} \breve{dk_j'}$$

$$\breve{dk_j'} = \frac{d^3k}{(2\pi)^3 2k^0}$$

↑ note that the VT from $\langle f|i\rangle^2$ was cancelled by V^2 from $\langle i|i\rangle$ T from $\dot{P}$

and $V^{n'}$ is cancelled by $\langle f|f\rangle$'s $V^{n'}$

Typically and in fact the initial particles are also in packs (not infinitely precise)

They come in flux.

The incident flux is the # of particles striking the other per unit volume.

1 Should read: per unit area per unit time (also in red below). Flux $= |\mathbf{k}_1|/(E_1 V)$, $[\text{flux}] = L^{-2}T^{-1}$, so $\sigma = \dot{P}/\text{flux}$ is an area.

~~We have~~ If we have 1 particle in volum V with a speed $v = |\vec{k}_1|/E_1$, that is striking particle 2 at rest.

We will have $|\vec{k}_1|/E_1/V$ particle 1 trying to strike particle 2 per unit time. per unit volume.

I can pack more particles in volume V so a more meaning full quantity is

$\dot{P}$ $[\dot{P}] = T^{-1}$

$\Leftarrow$ flux. $[\text{flux}] = L^{-2}T^{-1}$, which we call cross section.

quantifies how frequent two particles "meet"

Rate of meeting per unit time per unit volume

$$d\sigma = \frac{\dot{P}}{v\,|\vec{k}_1|/E_1/V} = \frac{(2\pi)^4 \delta^4(k_{in} - k_{out})}{4|\vec{k}_1|E_2} |T|^2 \prod_{j=1}^{n'} d\Pi_j$$

note that the above discussion was held in 2's rest frame

$|\vec{k}_1|_{F=T} E_2 = |\vec{k}_1|_{F=T} \cdot M_2$ it is in L.I. form

9.

1 Should read $+(m_1^2 - m_2^2)^2$ under the root, as on p. 2: $4|\mathbf{k}_1^{\rm FT}|m_2 = 2\sqrt{\lambda} = 4|\mathbf{k}_1^{\rm CM}|\sqrt{s}$ exactly, with $\lambda = s^2 - 2(m_1^2+m_2^2)s + (m_1^2-m_2^2)^2$. The page's $(m_1 - m_2)^2$ is wrong whenever $m_1 \neq m_2$.

~~flux~~ $\frac{1}{2}$

$$4|\vec{k}_1^{\rm FT}|m_2 = 2\sqrt{s^2 - 2(m_1^2+m_2^2)s + (m_1-m_2)^2}$$ (1)

And for typical $s \gg m_1^2, m_2^2$

$$4|\vec{k}_1^{\rm FT}|m_2 \cong 2s.$$

by comparing $|\vec{k}_1^{\rm FT}|$ and $|\vec{k}_1^{\rm cm}|$ we can see

$$4|\vec{k}_1^{\rm FT}|m_2 = 4|\vec{k}_1^{\rm cm}|\sqrt{s}$$

one gets a hint of L.I. for the flux factor here.

so $$d\sigma = \frac{1}{4|\vec{k}_1^{\rm cm}|\sqrt{s}}|\mathcal{T}|^2 d{\rm LIPS}_{n'}(k_1+k_2)$$

here

2 Filling in this heading: $4|\mathbf{k}_1^{\rm FT}|m_2 = 4|\mathbf{k}_1^{\rm CM}|\sqrt{s} = 4\sqrt{(k_1\cdot k_2)^2 - m_1^2 m_2^2}$, a scalar ($k_1\cdot k_2 = -E_1 m_2$ in the FT frame); $|\mathcal{T}|^2$ and dLIPS are scalars too, so $d\sigma$ is L.I. Physically, σ is an area transverse to the beams, and boosts along the beams leave transverse areas unchanged.

$$d{\rm LIPS}_{n'}(k_1+k_2) = (2\pi)^4 \delta^4(k_1+k_2 - k_{\rm out}) \prod_{j=1}^{n'} dk'_j$$

Comments on the L.I. of σ. (2)

~~So $2 \to n'$ pro~~

Now let's study a bit the behaviour of $d{\rm LIPS}_{n'}(k_1+k_2)$.

the most useful one is $\Delta\ n'=2$

In a similar fashion, we can see the $1 \to n'$ process would be.

$$\underbrace{\Gamma}_{\tau} = \frac{1}{2E_1} |\mathcal{T}|^2 \, dLIPS_{n'}(k_1).$$

decay rate, decay width

$d\Phi$ is L.I.

is $\mathcal{T}$ L.I.? (1)

for proper decay width

$$\Gamma_{1\to n'} = \frac{1}{2M_1} |\mathcal{T}|^2 dLIPS_{n'}$$

1 No. $\int |\mathcal{T}|^2 d\text{LIPS}$ is a scalar, a function of $k_1^2 = -M_1^2$ only, so all frame dependence sits in $1/2E_1 = 1/(2\gamma M_1)$: $\Gamma = \Gamma_{\text{rest}}/\gamma$, the time dilation of $\tau = 1/\Gamma$. The proper width (red) is the invariant.

Now let's see a bit of useful properties for

$$dLIPS_{n'}(k_{in} \to k_1' \cdots k_{n'}')$$

for $n' = 2$

$$dLIPS_2 = (2\pi)^4 \delta^4(k_{in} - k_1' - k_2') \, d\tilde{k}_1' \, d\tilde{k}_2'$$

It is Lorentz invariant, so we can work in any frame we want, let's choose c.m. frame.

$$dLIPS_2 = \frac{(2\pi)^4 \delta(\sqrt{s} - E_1' - E_2') \, d^3k_1'}{4(2\pi)^6 E_1' E_2'}$$

$$E_1' = \sqrt{\vec{k}_1'^2 + m_1'^2}, \quad E_2' = \sqrt{\vec{k}_2'^2 + m_2'^2} = \sqrt{\vec{k}_1'^2 + m_2'^2}$$

$$d^3\vec{k}_1' = |\vec{k}_1'^{cm}|^2 \, d|\vec{k}_1'^{cm}| \, d\Omega_{cm}$$

$$= |\vec{k}_1'^{cm}|^2 \, d|\vec{k}_1'^{cm}| \, d\cos\theta_{cm} \, d\phi_{cm}.$$

1 $S = 2$ goes into the total only: $\sigma = \frac{1}{2}\int_{4\pi} d\Omega\, d\sigma/d\Omega$ (or $\cos\theta > 0$ only), as (k_1', k_2') and (k_2', k_1') are one state. The $d\sigma/d\Omega$ above, without the $\frac{1}{2}$, is what a detector at θ counts.

(11)

Noting that $\int dx\, \delta(f(x)) = \sum_i |f'(x_i)|^{-1}$

$$\frac{2\,\delta(E_1' + E_2' - \sqrt{s})}{2|\vec{k}_1'^{cm}|} = \frac{|\vec{k}_1'|}{E_1} + \frac{|\vec{k}_1'^{cm}|}{E_2}$$

$$= |\vec{k}_1'^{cm}|\left(\frac{E_1' + E_2'}{E_1' E_2'}\right) = \frac{|\vec{k}_1'^{cm}|\sqrt{s}}{E_1 E_2}$$

$$d\mathrm{LIPS}_2(k) = \frac{|\vec{k}_1'^{cm}|}{16\pi^2 \sqrt{s}}\, d\Omega_{cm}$$

$$d\sigma = \frac{1}{64\pi^2 s} \frac{|\vec{k}_1'^{cm}|}{|\vec{k}_1^{cm}|} |\mathcal{T}|^2\, d\Omega_{cm}$$

another useful formula is

$$dt = 2|\vec{k}_1^{cm}||\vec{k}_1'^{cm}|\, d\cos\theta = 2|\vec{k}_1||\vec{k}_2|\frac{d\Omega_{cm}}{2\pi}$$

$$d\sigma = \frac{1}{64\pi s |\vec{k}_1^{cm}|^2} |\mathcal{T}|^2\, dt.$$ (This is L.I but $d\sigma/d\Omega$ is not since $d\Omega$ depends on frame)

L.I. with expressions earlier.

Finally, note that when population the final states, identical particles will "share" the phase space, so there will be a separate "symmetry factor" for $d\mathrm{LIPS}_n$, e.g. for two identical scalar, $\frac{1}{2!}$. that one needs to divide $S=2$

Added page 2025.

We've talked about the origin of unit Barn.

1 barn = 100 fm²

Proton size? $\mathcal{O}(\text{fm}) \rightarrow \pi - p$ $\sigma \sim \mathcal{O}(10\text{mb})$

inelastic ~ 60mb
single diffractive ~ 12mb
elastic ~ 40mb

$\sqrt{s} \simeq 10\,\text{TeV}$.

neutron U^{235} ~ 1 barn
for fast neutrons
$E > 100\,\text{keV}$.

The particle physics searches, e.g. Higgs discovery

$\sigma_{pp \to H \to \gamma\gamma} \simeq 50\,\text{pb} \times 10^{-3} \simeq 50\,\text{fb} = 50 \times 10^{-15}$ barn

compare with
100×10^{-3} barn

We discovery 10^{-12} rare rates
from the background.
(A needle in a haystack)

We actually measure stuff to 10^{-18} level.

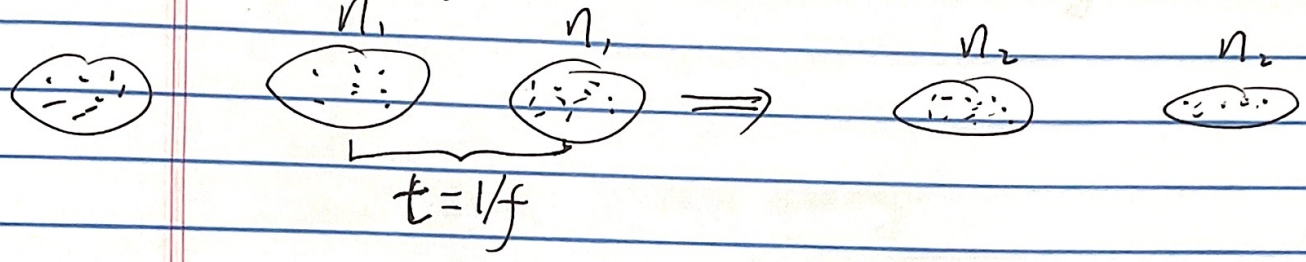
Cross section is 1 on 1 rate characterizes.

So what's rate?

A pairing concept is called "Luminosity"

Luminosity:

How intense the two spill of particles passing by each other.



$$\mathcal{L} = f n_1 n_2 / a.$$

particles / cm² / sec.

For instance:

LHC:

$$n_1 = n_2 = 10^{11}$$

$$a \simeq (64\,\mu m)^2$$

$$f = 1/(25 \text{ nano sec})$$

$$\mathcal{L} = 40 \times 10^6 \times 10^{22} \times \frac{1}{4000 \times (10^{-4})^2 \text{ cm}^2}$$

$$= 10^{34}/\text{cm}^2/\text{sec.}$$

$$= 10 \text{ nb}^{-1} \text{ sec}^{-1}$$

rate = 6.1 : $p-p$ $\quad 100 \text{ mb} \times 10 \text{ nb}^{-1} \text{ sec}^{-1}$

$$= 100 \times 10^6 \text{ nb} \times 10 \text{ nb}^{-1} \text{ sec}^{-1}$$

$$= 10^9/\text{sec.}$$

(We actually record about 10^3 event/sec.
that already provides 10 PB data/yr)

"big data"

One can see that

Cross Section & Luminosity is a "Pair"

↓ ↓

1 on 1 How I prepare my experiment.

⇓

Hence one can compare different experiments directly.

Phase space:

$$d\Phi_n(P; p_1 \dots p_n) = (2\pi)^4 \delta^{(4)}\left(P - \sum_{i=1}^{n} p_i\right) \prod_{i=1}^{n} \frac{d^3 p_i}{(2\pi)^3 2E_i}$$

and a recursive relation

$$d\Phi_n(P; p_1, \dots, p_n) = d\Phi_j(q; p_1 \dots p_j) \times d\Phi_{n-j+1}(P; q, p_{j+1} \dots p_n)(2\pi)^3 dq^2$$

What's the phase space volumn of n identical particles (that are massless)

$$\int dLIPS_n = \frac{\hat{S}^{n-2} (2\pi)^{-4}}{8\pi (16\pi^2)^{n-2} n!(n-1)!}$$

1 Should read

$$\frac{1}{8\pi\, n!} \frac{(\hat{s}/16\pi^2)^{n-2}}{(n-1)!\,(n-2)!}$$

for identical particles; the page lacks $(n-2)!$ (wrong for $n \geq 4$). Labels swap here: without the red $(2\pi)^{-4}$ this is the textbook LIPS_n ($n = 2$: $1/16\pi$, p. 11 with $S = 2$); with it, PDG's.

The red one is our textbook's dLIPS convention. the black one is PDG review convention.